S V V Sai Sri Vallabh Pataneni

Research Assistant/Associate (PhD): Fuel Cell Systems, PEM & SOFC

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SUMMARY

An Automotive Engineering Master's student at RWTH Aachen with over two years of hands-on PEM fuel-cell systems experience, from Germany's only student-built hydrogen fuel-cell vehicle (1 kW) to a 100 kW, 1.5 t FCEV pickup truck at Robert Bosch to a 150-200 kW aircraft powertrain at DLR. At DLR I sized and selected the system's balance-of-plant components, then designed and built its cathode-side control architecture, including a Model Predictive Control approach, in MATLAB/Simulink. Seeking a Research Assistant/Associate PhD position at RWTH's Chair of Thermodynamics of Mobile Energy Conversion Systems (TME) to bring this system-level modeling, control and test-bench experience to fuel-cell systems research for mobile applications.

KEY SKILLS

Fuel-cell systems & controls

PEM fuel-cell system architecture · 0D lumped-parameter plant modeling · 1D lumped modeling · feed-forward/PI/decoupler control design · Model Predictive Control (MPC) · back-pressure-valve (BPV) protection control · operating-strategy optimization · balance-of-plant sizing & selection · CAN communication

Modeling & simulation

MATLAB/Simulink · GT Suite · Dymola (Modelica) · Genetic Algorithms · Gradient Descent · DP/ECMS benchmarking

Test & measurement

fuel-cell test-bench build & commissioning · single-cell & stack testing · polarisation curves · purging/humidity cycling · measurement-data evaluation & troubleshooting

Programming & data

Python (NumPy, Pandas) · C/C++ · Power BI

Design & CFD

CATIA V5 · Siemens NX · ANSYS (Fluent, Structural)

PROFESSIONAL EXPERIENCE

Werkstudent, Fuel-Cell Systems (Aircraft), DLR Institute for Small Aircraft Technologies (INK) — 01/2026 – Present

Würselen (Aachen), Germany

- **Cathode-Side Control Architecture:** Built a 0D lumped-parameter plant model of the cathode loop in MATLAB/Simulink for the ~150-200 kW fuel-cell system powering EnginePod, DLR's hydrogen-electric powertrain for a 9-seater commuter aircraft, and designed feed-forward, PI and decoupler control plus back-pressure-valve (BPV) protection control on top of it.
- **Model Predictive Control:** Built a Model Predictive Control (MPC) approach for the same cathode-side control architecture, now tuning both control strategies' parameters against real system measurement data.
- **Higher-Fidelity Modeling:** Building a complementary 1D lumped plant model of the cathode system in GT Suite.
- **Balance-of-Plant Sizing & Component Selection:** Built Python- and Excel-based sizing tools to select balance-of-plant components, then derived design specifications and selected the cathode system, thermal-management system and full sensory-system components, driving their procurement.
- **System Rig Design:** Designed the complete fuel-cell system rig in CATIA V5 for packaging, piping, testing and component orientation, optimized for easy transferability and integration.

Mechanical / Suspension Team Lead & Pit Crew, Ecogenium e.V. (RWTH student team) — 06/2024 – Present

Aachen, Germany

- **Fuel-Cell System Testing & Troubleshooting:** Leading this season's fuel-cell system testing, integration and fault diagnosis on Germany's only student-built hydrogen fuel-cell vehicle, on its newly built water-cooled fuel-cell system, running structured test campaigns across the electrical, thermal and fluidic subsystems.
- **Energy-Management & Cooling System Design:** Designed this season's energy-management and cooling-system control strategy for the water-cooled fuel-cell system in MATLAB/Simulink.
- **Fuel-Cell Test Bench:** Engineered, fabricated and electronically integrated a standalone fuel-cell test bench for the prior air-cooled stack, running purging, humidity cycling, short-circuiting and polarisation-curve experiments and mapping its optimal thermal operating window.
- **End-to-End System Ownership:** Holistic ownership of fuel-cell system setup, from component sizing and selection through CAN-bus communication and control design to test and experimentation, gained across a runner-up finish at Shell Eco-Marathon 2025.

Operating Strategy / System Simulation Intern, Robert Bosch GmbH — 09/2025 – 12/2025

Schwieberdingen, Germany

- **Degradation-Safeguarding Strategy:** Engineered a MATLAB-based powertrain strategy for Strong Fuel Cell Hybrid EVs that safeguards fuel-cell and HV-battery degradation by managing dynamic loads, thermal stress, low-voltage operation, SOC swing and life cycles, parameterized via MAPs for different battery capacities and drive cycles.
- **Operating-Strategy Modelling:** Developed and validated an iterative Simulink model of a rule-based power-split operating strategy for a 100 kW-stack, 1.5 t FCEV pickup-truck platform in the (FC)EVsim PHEV module, correlating outputs against reference Excel and measurement data for high model fidelity.
- **Multi-Objective Optimization:** Executed multi-variable parametric optimization using Genetic Algorithms and Gradient Descent to generate control MAPs for charge sustenance and minimal fuel consumption, benchmarked against Dynamic Programming and ECMS.
- **Analysis & Documentation:** Built post-processing and KPI analysis (power distribution, SOC window, efficiency, degradation) and co-authored a peer-reviewed paper on the rule-based FCEV control strategy.

EDUCATION

M.Sc. Automotive Engineering, RWTH Aachen University — 09/2023 – present

Aachen, Germany · GPA 1.7 (1.2 in core automotive courses) · Upcoming Master's thesis at DLR, direction: fuel-cell system control (scope in definition) · Relevant modules: Alternative & Mobile Propulsion Systems, Simulation Sciences, Mechatronics, Thermodynamics, FEM

B.Tech Mechanical Engineering, JNTU Hyderabad — 06/2019 – 04/2023

Hyderabad, India · CGPA 9.05 (GPA 1.3, German scale)

PUBLICATIONS

- "A Comprehensive Analysis of Two Innovative Aircraft Design Configurations," IJSED Research, 11/2022.
- Co-author, peer-reviewed paper on a rule-based FCEV operating-strategy control approach (Robert Bosch GmbH), in review at SAE.

LANGUAGES

English (Fluent, C1) · German (A2-B1) · Telugu / Hindi (Native)